MATHEMATICS / INDEPENDENT STUDY

MM5020

Mathematics III - Abstract Algebra

Exercises and mock assessments

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 2 · 25 October 2026

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

1 Unit 1 · Groups and cyclic structure

MM5020-U1.01 · E · 15 min
Learning objective: U1.1 Group axioms
Show that the even integers form a group under addition, and explain why they do not form a group under multiplication.

MM5020-U1.02 · E · 15 min
Learning objective: U1.1 Cancellation
Solve $axb = c$ in an arbitrary group and prove uniqueness.

MM5020-U1.03 · E · 15 min
Learning objective: U1.2 Element orders
Find the order and generated subgroup of $[9]$ in $\mathbb{Z}/15\mathbb{Z}$.

MM5020-U1.04 · E · 15 min
Learning objective: U1.2 Subgroup test
Prove that the intersection of two subgroups of a group is a subgroup.

MM5020-U1.05 · C · 30 min
Learning objective: U1.1 Structural identities
Prove that a group in which $x^2 = e$ for every x is abelian.

MM5020-U1.06 · C · 30 min
Learning objective: U1.2 Cyclic subgroups
List all subgroups of $\mathbb{Z}/18\mathbb{Z}$ and their orders.

MM5020-U1.07 · C · 30 min
Learning objective: U1.2 Generator arithmetic
Let a have order n. Prove $\langle a^r \rangle = \langle a^{\gcd(n,r)} \rangle$.

MM5020-U1.08 · C · 30 min
Learning objective: U1.2 Subgroup structure
Prove that the union of two subgroups H, K is a subgroup exactly when one contains the other.

MM5020-U1.09 · C · 30 min
Learning objective: U1.1 Noncommutative examples
Let r rotate a square by a quarter turn and s reflect it in a vertical axis. Show $srs = r^{-1}$ and use this to prove the symmetry group is nonabelian.

MM5020-U1.10 · A · 50 min
Learning objective: U1.2 Automorphisms
Classify all automorphisms of the additive group $\mathbb{Z}/n\mathbb{Z}$.

MM5020-U1.11 · A · 50 min
Learning objective: U1.2 Products of cyclic groups
Prove that $C_m \times C_n$ is cyclic exactly when $\gcd(m, n) = 1$.

MM5020-U1.12 · A · 50 min
Learning objective: U1.1 Axioms and finiteness
Let M be a finite nonempty set with an associative operation and a two-sided identity. Assume left and right cancellation. Prove that M is a group.

MM5020-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: A disguised operation

On $X = \mathbb{R}\{-1\}$ define $a \star b = a + b + ab$. Determine whether this is a group, find its identity and inverses, and find all elements of finite order.

2 Unit 2 · Cosets and permutations

MM5020-U2.01 · E · 15 min
Learning objective: U2.1 Cosets
List the cosets of $H = \{[0], [4], [8]\}$ in $\mathbb{Z}/12\mathbb{Z}$.

MM5020-U2.02 · E · 15 min
Learning objective: U2.1 Lagrange
What element orders are possible in a group of order twenty-one according to Lagrange's theorem? Does this alone guarantee an element of order twenty-one?

MM5020-U2.03 · E · 15 min
Learning objective: U2.2 Permutation composition
Compute $(123)(23)$ and its sign, composing right to left.

MM5020-U2.04 · E · 15 min
Learning objective: U2.2 Cycle order
Find the order and sign of $(1234)(567)$ in S_7.

MM5020-U2.05 · C · 30 min
Learning objective: U2.1 Coset criteria
Prove $aH = bH$ if and only if $a^{-1}b \in H$.

MM5020-U2.06 · C · 30 min
Learning objective: U2.1 Prime order
Prove that a group of prime order has no proper nontrivial subgroup and is cyclic.

MM5020-U2.07 · C · 30 min
Learning objective: U2.2 Powers of cycles
Describe the disjoint cycles of $(123456)^2$ and $(123456)^3$, and compute their orders.

MM5020-U2.08 · C · 30 min
Learning objective: U2.2 Counting permutations
How many three-cycles are in S_5? How many products of two disjoint transpositions?

MM5020-U2.09 · C · 30 min
Learning objective: U2.2 Alternating groups
Show every three-cycle is even and every even permutation is a product of three-cycles.

MM5020-U2.10 · A · 50 min
Learning objective: U2.2 Minimal permutation degree
Find the least n for which S_n has an element of order fifteen.

MM5020-U2.11 · A · 50 min
Learning objective: U2.1 Indices in a tower
For finite G and $K \le H \le G$, prove $[G:K] = [G:H][H:K]$ by constructing coset representatives.

MM5020-U2.12 · A · 50 min
Learning objective: U2.2 Conjugation and cycles
Prove $\tau(a_1 \dots a_r)\tau^{-1} = (\tau(a_1) \dots \tau(a_r))$. Deduce that conjugate permutations have the same cycle lengths, and conversely.

MM5020-U2.13 · A · 60 min · Optional challenge

Learning objective: U2 Transfer: Symmetries with fixed positions

How many permutations of six letters have order two and exactly two fixed letters? Prove that any two are conjugate, and determine how many permutations commute with one of them.

3 Unit 3 · Homomorphisms and quotients

MM5020-U3.01 · E · 15 min
Learning objective: U3.1 Kernels
Find the kernel and image of $f : \mathbb{Z} \to \mathbb{Z}/8\mathbb{Z}$, $f(n) = [3n]$.

MM5020-U3.02 · E · 15 min
Learning objective: U3.1 Normality
Show the kernel of the determinant map $\mathrm{GL}_2(\mathbb{R}) \to \mathbb{R}^\times$ is normal.

MM5020-U3.03 · E · 15 min
Learning objective: U3.2 Quotient size
Let $|G| = 60$ and $N \lhd G$ have order twelve. Determine $|G/N|$ and its group structure.

MM5020-U3.04 · E · 15 min
Learning objective: U3.1 Homomorphisms
Show a homomorphism $f : G \to H$ satisfies $f(g^n) = f(g)^n$ for every integer n.

MM5020-U3.05 · C · 30 min
Learning objective: U3.2 First isomorphism theorem
Identify $\mathbb{R}^\times/\{1, -1\}$ using a natural homomorphism.

MM5020-U3.06 · C · 30 min
Learning objective: U3.1 Normality
Prove the center $Z(G)$ is a normal subgroup.

MM5020-U3.07 · C · 30 min
Learning objective: U3.2 Quotient maps
For positive integers m, n, when is $[a]_m \mapsto [a]_n$ well defined? When it is, find its kernel.

MM5020-U3.08 · C · 30 min
Learning objective: U3.2 Correspondence
List all subgroups of $\mathbb{Z}/12\mathbb{Z}$ by using subgroups of $\mathbb{Z}$ containing $12\mathbb{Z}$.

MM5020-U3.09 · C · 30 min
Learning objective: U3.1 Cyclic quotient by center
Prove that if $G/Z(G)$ is cyclic, then G is abelian.

MM5020-U3.10 · A · 50 min
Learning objective: U3.2 Universal property
Prove that a homomorphism $f : G \to H$ factors through G/N exactly when $N \subseteq \ker f$, and prove uniqueness of the factor.

MM5020-U3.11 · A · 50 min
Learning objective: U3.2 Abelianization
Let $[G, G]$ be generated by all commutators $aba^{-1}b^{-1}$. Show it is normal, its quotient is abelian, and every map from G to an abelian group factors uniquely through this quotient.

MM5020-U3.12 · A · 50 min
Learning objective: U3.1 Failure of converse to Lagrange
Prove that A_4 has no subgroup of order six. You may use that three-cycles generate A_4.

MM5020-U3.13 · A · 60 min · Optional challenge

Learning objective: U3 Transfer: A large cyclic kernel

Let $G = (\mathbb{Z}/12\mathbb{Z}) \times (\mathbb{Z}/18\mathbb{Z})$. Define $f : G \to \mathbb{Z}/6\mathbb{Z}$ by $f(a, b) = a + b$. Determine the kernel and prove that it is cyclic; give a generator and the quotient up to isomorphism.

4 Unit 4 · Actions and simple groups

MM5020-U4.01 · E · 15 min
Learning objective: U4.1 Actions
Verify that S_n acts on $\{1, \dots, n\}$ by evaluation.

MM5020-U4.02 · E · 15 min
Learning objective: U4.1 Stabilizers
Find the stabilizer of one in the natural S_4 action and its order.

MM5020-U4.03 · E · 15 min
Learning objective: U4.2 Conjugacy
Compute $(123)(12)(123)^{-1}$.

MM5020-U4.04 · E · 15 min
Learning objective: U4.2 Simplicity
Which nontrivial finite cyclic groups are simple?

MM5020-U4.05 · C · 30 min
Learning objective: U4.1 Kernel of an action
Show the kernel of the homomorphism associated with an action is the intersection of all stabilizers.

MM5020-U4.06 · C · 30 min
Learning objective: U4.1 Conjugate stabilizers
Prove $G_{h \cdot x} = hG_x h^{-1}$.

MM5020-U4.07 · C · 30 min
Learning objective: U4.2 Class equation
Use conjugacy classes to show the center of S_3 is trivial.

MM5020-U4.08 · C · 30 min
Learning objective: U4.2 Prime-power groups
Prove a group of order p^2 is abelian.

MM5020-U4.09 · C · 30 min
Learning objective: U4.1 Orbit counting
Count two-colorings of four labeled square vertices up to rotation, with reflections excluded.

MM5020-U4.10 · A · 50 min
Learning objective: U4.2 Simple prime-power groups
Classify finite simple groups whose order is a power of a prime.

MM5020-U4.11 · A · 50 min
Learning objective: U4.1 Coset actions
Let H have finite index n in G. Show there is a normal subgroup $N \subseteq H$ with $[G : N] \leq n!$.

MM5020-U4.12 · A · 50 min
Learning objective: U4.2 Faithful actions of simple groups
Prove every nontrivial action of a simple group is faithful. Deduce that A_5 cannot act nontrivially on a set of four elements.

MM5020-U4.13 · A · 60 min · Optional challenge

Learning objective: U4 Transfer: Four-position ornaments

An ornament has four positions at the vertices of a square, each colored with one of three colors. Two colorings are identified if a rotation or reflection of the square relates them. How many ornaments are there? Explain why simply dividing the number of colorings by eight fails.

5 Unit 5 · Rings and their quotients

MM5020-U5.01 · E · 15 min
Learning objective: U5.1 Units
Find every unit in $\mathbb{Z}/10\mathbb{Z}$ and its inverse.

MM5020-U5.02 · E · 15 min
Learning objective: U5.1 Zero divisors
Give nonzero matrices $A, B \in M_2(\mathbb{Q})$ with $AB = 0$.

MM5020-U5.03 · E · 15 min
Learning objective: U5.2 Ideals
Show $6\mathbb{Z}$ is an ideal of $\mathbb{Z}$, and describe the identity of its quotient.

MM5020-U5.04 · E · 15 min
Learning objective: U5.1 Quaternions
Find the inverse of $q = 1 + 2i - j$.

MM5020-U5.05 · C · 30 min
Learning objective: U5.2 Quotient criteria
Determine whether (5) and (6) are prime or maximal in $\mathbb{Z}$.

MM5020-U5.06 · C · 30 min
Learning objective: U5.2 Evaluation
Show $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$.

MM5020-U5.07 · C · 30 min
Learning objective: U5.1 Characteristic
Prove a nonzero domain has characteristic zero or a prime.

MM5020-U5.08 · C · 30 min
Learning objective: U5.2 Ideals and units
Prove an ideal contains a unit if and only if it is the whole ring. Deduce that a field has exactly two ideals.

MM5020-U5.09 · C · 30 min
Learning objective: U5.2 Noncommutative ideals
Show the matrices with zero second column form a left ideal of $M_2(K)$ but not a two-sided ideal.

MM5020-U5.10 · A · 50 min
Learning objective: U5.2 Chinese remainder theorem
Prove $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ for coprime positive m, n.

MM5020-U5.11 · A · 50 min
Learning objective: U5.2 Matrix ring ideals
Prove every two-sided ideal of $M_{n(K)}$, for a field K, is zero or the whole ring.

MM5020-U5.12 · A · 50 min
Learning objective: U5.1 Division rings
Prove quaternion conjugation reverses products and deduce multiplicativity of $N(q) = q\overline{q}$.

MM5020-U5.13 · A · 60 min · Optional challenge
Learning objective: U5 Transfer: Coordinates hidden in residues
Find all e in $\mathbb{Z}/12\mathbb{Z}$ satisfying $e^2 = e$. Construct an explicit ring isomorphism from $\mathbb{Z}/12\mathbb{Z}$ to $(\mathbb{Z}/3\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z})$ and use it to explain your list.

6 Unit 6 · Polynomials and factorization

MM5020-U6.01 · E · 15 min
Learning objective: U6.1 Division
Divide $x^3 + 2x + 1$ by $x^2 + 1$ over $\mathbb{Q}$.

MM5020-U6.02 · E · 15 min
Learning objective: U6.1 Bézout
Find integers u, v with $30u + 21v = 3$.

MM5020-U6.03 · E · 15 min
Learning objective: U6.2 Irreducibility
Prove $x^3 - 5$ is irreducible over $\mathbb{Q}$.

MM5020-U6.04 · E · 15 min
Learning objective: U6.1 Roots
Factor $x^2 + 1$ over $\mathbb{F}_5$.

MM5020-U6.05 · C · 30 min
Learning objective: U6.1 Modular inverses
Find the inverse of $x + 1$ in $\mathbb{Q}[x]/(x^2 + 1)$.

MM5020-U6.06 · C · 30 min
Learning objective: U6.2 Reduction test
Prove $x^3 + x + 1$ is irreducible over $\mathbb{Q}$ by reducing modulo two.

MM5020-U6.07 · C · 30 min
Learning objective: U6.2 Failure outside PIDs
Show 2 is irreducible but not prime in $\mathbb{Z}\left[\sqrt{-5}\right]$.

MM5020-U6.08 · C · 30 min
Learning objective: U6.1 Principal ideals
Show $K[x]/(f)$ is a field if and only if f is irreducible, assuming f is nonconstant.

MM5020-U6.09 · C · 30 min
Learning objective: U6.2 Primitive products
Show the product of two primitive integer polynomials is primitive.

MM5020-U6.10 · A · 50 min
Learning objective: U6.2 Shifted Eisenstein
For prime p, prove $1 + x + \ldots + x^{p-1}$ is irreducible over $\mathbb{Q}$.

MM5020-U6.11 · A · 50 min
Learning objective: U6.1 Polynomial Chinese remainder
If $f, g \in K[x]$ are coprime, prove $K[x]/(fg) \cong K[x]/(f) \times K[x]/(g)$.

MM5020-U6.12 · A · 50 min
Learning objective: U6.2 Nonprincipal ideals
Prove $\mathbb{Z}[x]$ is a UFD but not a PID.

MM5020-U6.13 · A · 60 min · Optional challenge

Learning objective: U6 Transfer: Two quartics without rational roots

Determine which of $x^4 + 1$ and $x^4 + 4$ are irreducible over $\mathbb{Q}$. Give complete factorizations into irreducibles in $\mathbb{Z}[x]$ and explain why testing rational roots does not settle either question.

7 Unit 7 · Fields as vector spaces

MM5020-U7.01 · E · 15 min
Learning objective: U7.1 Fractions
Prove $\frac{a}{b} = 0$ in the fraction field of a domain exactly when $a = 0$.

MM5020-U7.02 · E · 15 min
Learning objective: U7.2 Minimal polynomials
Find the minimal polynomial of $\sqrt{7}$ over $\mathbb{Q}$ and a basis of its generated field.

MM5020-U7.03 · E · 15 min
Learning objective: U7.1 Tower law
If $[L : K] = 3$ and $[M : L] = 4$, find $[M : K]$. Can an intermediate field between K and M have degree five over K?

MM5020-U7.04 · E · 15 min
Learning objective: U7.2 Power-basis arithmetic
With $\alpha^3 = 2$, simplify $\alpha^5 + \alpha^4$ in the power basis.

MM5020-U7.05 · C · 30 min
Learning objective: U7.2 Inverses
Find $\left(2 + \sqrt{3}\right)^{-1}$ in $\mathbb{Q}\left(\sqrt{3}\right)$ and explain why every nonzero $a + b\sqrt{3}$ has an inverse in this field.

MM5020-U7.06 · C · 30 min
Learning objective: U7.2 Algebraicity
Prove every element of a finite extension is algebraic.

MM5020-U7.07 · C · 30 min
Learning objective: U7.2 Change of generator
Find the minimal polynomial of $1+\sqrt{2}$ over $\mathbb{Q}$.

MM5020-U7.08 · C · 30 min
Learning objective: U7.1 Fraction universal property
Explain why the reduction map $\mathbb{Z} \to \mathbb{F}_5$ cannot extend to a unital homomorphism $\mathbb{Q} \to \mathbb{F}_5$.

MM5020-U7.09 · C · 30 min
Learning objective: U7.2 Root adjunction
Construct a field containing a root of x^3+x+1 over $\mathbb{F}_2$ and list a basis over $\mathbb{F}_2$.

MM5020-U7.10 · A · 50 min
Learning objective: U7.1 Compositum degree
Prove $\left[\mathbb{Q}\left(\sqrt{2},\sqrt{5}\right):\mathbb{Q}\right]=4$ and find a basis.

MM5020-U7.11 · A · 50 min
Learning objective: U7.2 Transitivity of algebraicity
Let L/K be algebraic and let α be algebraic over L. Prove α is algebraic over K, even when L/K is infinite.

MM5020-U7.12 · A · 50 min
Learning objective: U7.2 A primitive generator
Show $\mathbb{Q}\left(\sqrt{2}+\sqrt{3}\right)=\mathbb{Q}\left(\sqrt{2},\sqrt{3}\right)$ and find the minimal polynomial of the sum.

MM5020-U7.13 · A · 60 min · Optional challenge

Learning objective: U7 Transfer: A field containing three roots

Let a be the positive real cube root of two and let ω be a nonreal root of $x^2 + x + 1$. Determine the smallest subfield of $\mathbb{C}$ containing every root of $x^3 - 2$, and its degree over $\mathbb{Q}$.

8 Unit 8 · Finite fields and applications

MM5020-U8.01 · E · 15 min
Learning objective: U8.1 Field orders
Which of the orders $8, 9, 10, 16, 18, 25$ can occur for a finite field?

MM5020-U8.02 · E · 15 min
Learning objective: U8.1 Arithmetic in four elements
In $\mathbb{F}_4 = \mathbb{F}_2[\alpha]$ with $\alpha^2 + \alpha + 1 = 0$, compute $\alpha(\alpha + 1)$ and the inverse of $\alpha + 1$.

MM5020-U8.03 · E · 15 min
Learning objective: U8.2 Generators
Find the order of two in $\mathbb{F}_7^{\times}$.

MM5020-U8.04 · E · 15 min
Learning objective: U8.1 Characteristic and dimension
What are the characteristic and dimension over the prime field of a field of order eighty-one?

MM5020-U8.05 · C · 30 min
Learning objective: U8.2 Frobenius
Prove the map $a \mapsto a^p$ is an automorphism of a finite field of characteristic p.

MM5020-U8.06 · C · 30 min
Learning objective: U8.2 Interpolation
Find the degree-less-than-three polynomial over $\mathbb{F}_5$ with $f(0) = 1$, $f(1) = 0$, $f(2) = 2$.

MM5020-U8.07 · C · 30 min
Learning objective: U8.1 Polynomial roots
Show every element of $\mathbb{F}_q$ is a root of $x^q - x$, and this polynomial has no repeated root.

MM5020-U8.08 · C · 30 min
Learning objective: U8.2 Evaluation codes
Evaluate all degree-less-than-two polynomials at the five elements of $\mathbb{F}_5$. How many codewords result, and what is their minimum distance?

MM5020-U8.09 · C · 30 min
Learning objective: U8.2 Primitive elements
Find all generators of $\mathbb{F}_7^\times$, given that three is one generator.

MM5020-U8.10 · A · 50 min
Learning objective: U8.1 Subfields
Show a field of sixteen elements has a unique subfield of four elements.

MM5020-U8.11 · A · 50 min
Learning objective: U8.2 Polynomial functions
Prove every function $\mathbb{F}_q \to \mathbb{F}_q$ is represented by a unique polynomial of degree less than q. Is the representing polynomial unique without the degree bound?

MM5020-U8.12 · A · 50 min
Learning objective: U8.1 Irreducible quadratics
Count the monic irreducible quadratics over $\mathbb{F}_q$.

MM5020-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: An equation over a larger finite field

For each $b \in \mathbb{F}_q$, determine the number of $x \in \mathbb{F}_{q^2}$ satisfying $x^q + x = b$. Include the case of characteristic two.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M1.01 · E · 15 min
Learning objective: U1–U2 Cyclic groups
List the orders of every element of $\mathbb{Z}/8\mathbb{Z}$.

MM5020-M1.02 · E · 15 min
Learning objective: U5 Ring quotients
Determine whether $\mathbb{Z}/15\mathbb{Z}$ is a field and list its units.

MM5020-M1.03 · C · 30 min
Learning objective: U3–U4 Group structure
Let $|G| = 49$. Prove G is abelian, and show that if some element has order forty-nine then G is cyclic.

MM5020-M1.04 · C · 30 min
Learning objective: U6–U7 Algebraic extensions
Let $\alpha^3 = 3$ over $\mathbb{Q}$. Prove the extension degree is three and find $(\alpha - 1)^{-1}$.

MM5020-M1.05 · A · 40 min
Learning objective: U3 Universal constructions
For normal subgroups N, M of G, show $g \mapsto (gN, gM)$ induces an injective homomorphism $G/(N \cap M) \to G/N \times G/M$. When is it surjective?

MM5020-M1.06 · A · 40 min
Learning objective: U8 Field structure
Prove that the fixed field of Frobenius $x \mapsto x^p$ in $\mathbb{F}_{p^n}$ is exactly $\mathbb{F}_p$.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M2.01 · E · 15 min
Learning objective: U2 Permutations
Find the order and sign of $(123)(45)(67)$.

MM5020-M2.02 · E · 15 min
Learning objective: U7 Minimal polynomials
Find the minimal polynomial of $2+\sqrt{5}$ over $\mathbb{Q}$.

MM5020-M2.03 · C · 30 min
Learning objective: U3 Quotient isomorphisms
Prove $S_4/A_4 \cong C_2$.

MM5020-M2.04 · C · 30 min
Learning objective: U5–U6 Field construction
Prove $\mathbb{F}_3[x]/(x^2+1)$ is a field and find the inverse of $[x+1]$.

MM5020-M2.05 · A · 40 min
Learning objective: U4 Actions
A group G of order fifteen acts on a set of seven elements. Prove there is a fixed point.

MM5020-M2.06 · A · 40 min
Learning objective: U7–U8 Degrees and subfields
Prove that $\mathbb{F}_{p^3}$ has no subfield strictly between $\mathbb{F}_p$ and itself.